

```

mysql> INSERT INTO class_info VALUES
      -> (7,'NOIDA'),
      -> (8,'PANIPAT');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

```

```

mysql>
mysql> SELECT *
      -> FROM class
      -> LEFT OUTER JOIN class_info
      -> ON class.id=class_info.id;

```

id	name	id	address
1	abhi	1	DELHI
1	abhi	1	DELHI
2	adam	2	MUMBAI
2	adam	2	MUMBAI
4	alex	NULL	NULL
1	abhi	1	DELHI
1	abhi	1	DELHI
2	adam	2	MUMBAI
2	adam	2	MUMBAI
3	alex	3	CHENNAI
3	alex	3	CHENNAI
4	anu	NULL	NULL
5	ashish	NULL	NULL

13 rows in set (0.00 sec)

```

mysql> SELECT COUNT(*)
      -> FROM
      -> (
      -> SELECT * FROM first_table
      -> UNION ALL
      -> SELECT * FROM second_table
      -> ) A;

```

COUNT(*)
4

1 row in set (0.00 sec)

```
MySQL 8.0 Command Line Cli  ×  +  ∨

mysql> SELECT class.name,
->         class_info.address
-> FROM class
-> INNER JOIN class_info
-> ON class.id=class_info.id;

+-----+-----+
| name | address |
+-----+-----+
| abhi | DELHI   |
| abhi | DELHI   |
| adam | MUMBAI  |
| adam | MUMBAI  |
| abhi | DELHI   |
| abhi | DELHI   |
| adam | MUMBAI  |
| adam | MUMBAI  |
| alex | CHENNAI |
| alex | CHENNAI |
+-----+-----+
10 rows in set (0.00 sec)
```

```
MySQL 8.0 Command Line Cli  X  +  v

mysql> SELECT c.id, c.name, ci.address
-> FROM class c
-> INNER JOIN class_info ci
-> ON c.id = ci.id
-> WHERE c.name = 'adam';
+-----+-----+-----+
| id  | name | address |
+-----+-----+-----+
| 2   | adam | MUMBAI  |
| 2   | adam | MUMBAI  |
| 2   | adam | MUMBAI  |
| 2   | adam | MUMBAI  |
+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> SELECT DISTINCT f.name
-> FROM first_table f
-> INNER JOIN second_table s
-> ON f.name = s.name;
+-----+
| name |
+-----+
| adam |
+-----+
1 row in set (0.00 sec)
```

```
MySQL 8.0 Command Line Cli  X  +  v

mysql> SELECT *
-> FROM class
-> RIGHT JOIN class_info
-> ON class.id=class_info.id
-> WHERE class.id IS NULL;
+-----+-----+-----+-----+
| id  | name | id  | address |
+-----+-----+-----+-----+
| NULL | NULL | 7   | NOIDA   |
| NULL | NULL | 8   | PANIPAT |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> SELECT f.name
-> FROM first_table f
-> LEFT JOIN second_table s
-> ON f.name = s.name
-> WHERE s.name IS NULL;
+-----+
| name |
+-----+
| abhi |
+-----+
1 row in set (0.00 sec)
```

```
MySQL 8.0 Command Line Cli  X  +  v

mysql> SELECT *
-> FROM class
-> NATURAL JOIN class_info;
+-----+-----+-----+
| id  | name | address |
+-----+-----+-----+
| 1  | abhi | DELHI   |
| 1  | abhi | DELHI   |
| 2  | adam | MUMBAI  |
| 2  | adam | MUMBAI  |
| 1  | abhi | DELHI   |
| 1  | abhi | DELHI   |
| 2  | adam | MUMBAI  |
| 2  | adam | MUMBAI  |
| 3  | alex | CHENNAI |
| 3  | alex | CHENNAI |
+-----+-----+-----+
10 rows in set (0.00 sec)

mysql>
mysql> INSERT INTO class VALUES
-> (5, 'ashish');
Query OK, 1 row affected (0.01 sec)
```

```
MySQL 8.0 Command Line Cli  X  +  v

mysql> SELECT *
      -> FROM class
      -> CROSS JOIN class_infor;
ERROR 1146 (42S02): Table 'joins.class_infor' doesn't exist
mysql>
mysql> SELECT *
      -> FROM class
      -> CROSS JOIN class_info;
+-----+-----+-----+-----+
| id  | name | id  | address |
+-----+-----+-----+-----+
| 1  | abhi | 1  | DELHI   |
| 2  | adam | 1  | DELHI   |
| 4  | alex | 1  | DELHI   |
| 1  | abhi | 2  | MUMBAI  |
| 2  | adam | 2  | MUMBAI  |
| 4  | alex | 2  | MUMBAI  |
| 1  | abhi | 3  | CHENNAI |
| 2  | adam | 3  | CHENNAI |
| 4  | alex | 3  | CHENNAI |
+-----+-----+-----+-----+
9 rows in set (0.00 sec)

mysql>
mysql> CREATE TABLE class (
      ->     id INT,
      ->     name VARCHAR(30)
      -> );
ERROR 1050 (42S01): Table 'class' already exists
mysql> INSERT INTO class VALUES
      -> (1,'abhi'),
      -> (2,'adam'),
      -> (3,'alex'),
      -> (4,'anu');
Query OK, 4 rows affected (0.01 sec)
```

```
MySQL 8.0 Command Line Cli  x  +  v

mysql>
mysql> SELECT *
      -> FROM class
      -> LEFT JOIN class_info
      -> ON class.id=class_info.id
      -> WHERE class_info.id IS NULL;
+-----+-----+-----+-----+
| id  | name  | id  | address |
+-----+-----+-----+-----+
| 4   | alex  | NULL | NULL    |
| 4   | anu   | NULL | NULL    |
| 5   | ashish | NULL | NULL    |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> SELECT f.*
      -> FROM first_table f
      -> LEFT JOIN second_table s
      -> ON f.name = s.name
      -> WHERE s.name IS NULL;
+-----+-----+
| id  | name |
+-----+-----+
| 1   | abhi |
+-----+-----+
1 row in set (0.00 sec)
```

```
MySQL 8.0 Command Line Cli  ×  +  v

mysql>
mysql> CREATE TABLE first_table(
  -> id INT,
  -> name VARCHAR(30)
  -> );
Query OK, 0 rows affected (0.08 sec)

mysql>
mysql> CREATE TABLE second_table(
  -> id INT,
  -> name VARCHAR(30)
  -> );
Query OK, 0 rows affected (0.05 sec)

mysql> INSERT INTO first_table VALUES
  -> (1,'abhi'),
  -> (2,'adam');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO second_table VALUES
  -> (2,'adam'),
  -> (3,'chester');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM first_table
  -> UNION
  -> SELECT * FROM second_table;
```

```
mysql> SELECT a.*
  -> FROM first_table a
  -> INNER JOIN second_table b
  -> ON a.id = b.id;
+-----+-----+
| id  | name |
+-----+-----+
|  2  | adam |
+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> create database joins;
Query OK, 1 row affected (0.01 sec)

mysql> use joins;
Database changed
mysql> CREATE TABLE class (
  ->     id INT,
  ->     name VARCHAR(30)
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql> CREATE TABLE class_info (
  ->     id INT,
  ->     address VARCHAR(30)
  -> );
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> INSERT INTO class VALUES
  -> (1, 'abhi'),
  -> (2, 'adam'),
  -> (4, 'alex');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO class_info VALUES
  -> (1, 'DELHI'),
  -> (2, 'MUMBAI'),
  -> (3, 'CHENNAI');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
```

```
MySQL 8.0 Command Line Cli  ×  +  ∨

Records: 4  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO class_inform VALUES
  -> (1, 'DELHI'),
  -> (2, 'MUMBAI'),
  -> (3, 'CHENNAI');
ERROR 1146 (42S02): Table 'joins.class_inform' doesn't exist
mysql> INSERT INTO class_info VALUES
  -> (1, 'DELHI'),
  -> (2, 'MUMBAI'),
  -> (3, 'CHENNAI');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

id	name
1	abhi
2	adam
3	chester

3 rows in set (0.00 sec)

mysql>

mysql> SELECT name FROM first_table

-> UNION

-> SELECT name FROM second_table;

name
abhi
adam
chester

3 rows in set (0.00 sec)

mysql> SELECT * FROM first_table

-> UNION ALL

-> SELECT * FROM second_table;

id	name
1	abhi
2	adam
2	adam
3	chester

```
MySQL 8.0 Command Line Cli  ×  +  ∨

mysql> SELECT c.id, c.name, ci.address
-> FROM class c
-> INNER JOIN class_info ci
-> ON c.id = ci.id
-> WHERE ci.address IS NOT NULL;
```

id	name	address
1	abhi	DELHI
1	abhi	DELHI
2	adam	MUMBAI
2	adam	MUMBAI
1	abhi	DELHI
1	abhi	DELHI
2	adam	MUMBAI
2	adam	MUMBAI
3	alex	CHENNAI
3	alex	CHENNAI

10 rows in set (0.00 sec)

```
mysql> SELECT c.id, c.name, ci.address
-> FROM class c
-> LEFT JOIN class_info ci
-> ON c.id = ci.id
-> WHERE ci.address IS NULL;
```

id	name	address
4	alex	NULL
4	anu	NULL
5	ashish	NULL

3 rows in set (0.00 sec)

```
mysql> SELECT COUNT(*)  
-> FROM  
-> (  
-> SELECT * FROM first_table  
-> UNION ALL  
-> SELECT * FROM second_table  
-> ) A;
```

```
+-----+  
| COUNT(*) |  
+-----+  
|         4 |  
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> SELECT c.id,
->         c.name,
->         CASE
->             WHEN ci.address IS NULL THEN 'Address Missing'
->             ELSE 'Address Available'
->         END AS Status
-> FROM class c
-> LEFT JOIN class_info ci
-> ON c.id = ci.id;
```

id	name	Status
1	abhi	Address Available
1	abhi	Address Available
2	adam	Address Available
2	adam	Address Available
4	alex	Address Missing
1	abhi	Address Available
1	abhi	Address Available
2	adam	Address Available
2	adam	Address Available
3	alex	Address Available
3	alex	Address Available
4	anu	Address Missing
5	ashish	Address Missing

13 rows in set (0.00 sec)

```
mysql> SELECT c.id, c.name, ci.address
-> FROM class c
-> INNER JOIN class_info ci
-> ON c.id = ci.id
-> WHERE ci.address IS NOT NULL;
```

c Syntax to use here: SELECT name FROM second_table as line 5

```
mysql> SELECT DISTINCT f.name
-> FROM first_table f
-> INNER JOIN second_table s
-> ON f.name = s.name;
```

name
adam

1 row in set (0.00 sec)

```
mysql> SELECT f.*
```

JOINS_OPERATIONS_DB_11

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```
mysql> SELECT f.*
-> FROM first_table f
-> LEFT JOIN second_table s
-> ON f.name = s.name
-> WHERE s.name IS NULL;
+-----+-----+
| id  | name |
+-----+-----+
| 1   | abhi |
+-----+-----+
1 row in set (0.00 sec)

mysql>
```

```
MySQL 8.0 Command Line Cli  ×  +  ▾

mysql> SELECT *
-> FROM class
-> RIGHT OUTER JOIN class_info
-> ON class.id=class_info.id;
+-----+-----+-----+-----+
| id  | name | id  | address |
+-----+-----+-----+-----+
| 1   | abhi | 1   | DELHI   |
| 1   | abhi | 1   | DELHI   |
| 2   | adam | 2   | MUMBAI  |
| 2   | adam | 2   | MUMBAI  |
| 3   | alex | 3   | CHENNAI |
| 1   | abhi | 1   | DELHI   |
| 1   | abhi | 1   | DELHI   |
| 2   | adam | 2   | MUMBAI  |
| 2   | adam | 2   | MUMBAI  |
| 3   | alex | 3   | CHENNAI |
| NULL | NULL | 7   | NOIDA   |
| NULL | NULL | 8   | PANIPAT |
+-----+-----+-----+-----+
12 rows in set (0.00 sec)
```